

	<i>Attack direction</i>	<i>Per-edge rank</i>	<i>Per-node radius</i>	<i>Realistic edge-only</i>
Nettack / Mettack [1, 2]		• surrogate grad		
GR-BCD [3]	• block-coord	• greedy ranks		
Localized smoothing [4]			• probabilistic	
AGNNCert [5]			• deterministic	
AEGIS (<i>ours</i>)	• closed-form SVD	• $\ [S_c]_{:,k} \ _2$	• first-order r_v	• S_c matrix

Attacks (•) surface attack-direction or per-edge information on surrogate losses; **certifiers** (•) surface per-node radii but no attack direction or rank; **AEGIS** (•) surfaces all four from a *single* closed-form computation of S_c under realistic (symmetric, edge-only) perturbations. Citations [1]–[5] resolve to the paper’s reference list.